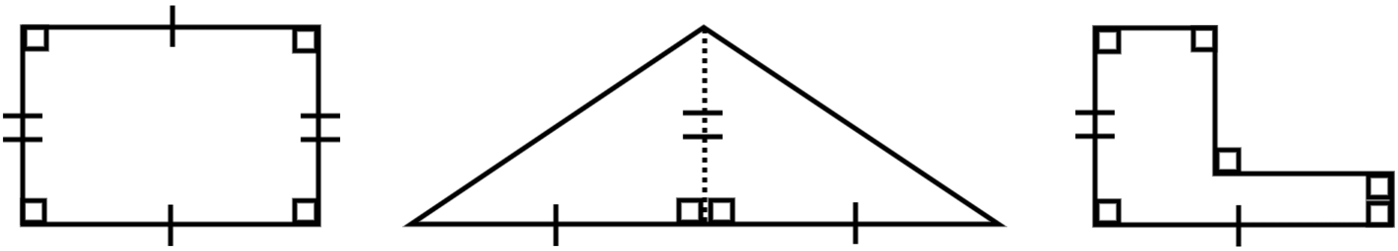
Area of composite rectilinear shapes

Task A: Comparing areas

1) Order the shapes from smallest to the largest area.

a) b) c)



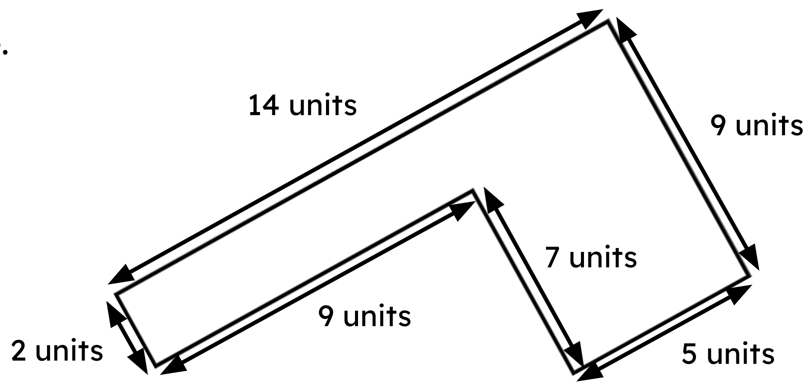
C is the smallest as it fits inside A with space left over.
B can be split into two triangles along the dotted line and these can be rearrange to make A. Therefore B and A have the same area.

Task B: Three methods to find the area

1) Calculate the area of this composite rectilinear shape using each of the three methods.

Which is most efficient?

Justify your choice.



E.g.

Area = $2 \times 9 + 5 \times 9$	*Area = $14 \times 9 - 7 \times 9$*	*Area = $(2 + 5) \times 9$*
Area = $18 + 45$	*Area = $126 - 63$*	*Area = 7×9*
Area = 63 units^2	*Area = 63 units^2*	*Area = 63 units^2*
E.g. I think this method is the most efficient as it is the one I find the easiest to do. This method is easy to always do.	*E.g. I think this method is the most efficient as it involves less work when the composite rectilinear shape is more complicated.*	*E.g. I think this method is the most efficient as it made just one rectangle that I had to find the area of.*

Task C: Evaluating the methods

1) Draw your own composite rectilinear shape and try to find the area using the three methods.

Can you draw a shape where one (or more) of these methods would not be helpful?

Shapes will vary but should all be built from rectangles placed adjacent.

Here is an example of a shape where completing and rearranging are not as helpful.